

Dear Dr. de Ruijter-Villani and Dr. Claes,

My name is Varvara Lazarenko, and I would like to express my strong interest in the PhD position in Aneuploidy in Mammalian Embryos / Clinical Veterinary Embryologist. The combination of translational embryology, reproductive genetics, and clinical application speaks directly to my academic background, hands-on research experience, and personal aspirations. This unique position aligns with my long-standing interest in biology and my passion for equine sciences.

Throughout my education, I have received a strong academic foundation and training in laboratory techniques central to molecular biology, genome analysis, and physiology. During my Bachelor's at Lomonosov Moscow State University (MSU), I built a solid skillset in wet lab methods, including PCR, gel electrophoresis, DNA/RNA extraction, ELISA, microscopy, immunohistochemistry, intracellular recordings, and behavioural testing in rodent models. My practical experience in physiology and molecular biology was deepened during my year-long role as a Junior Research Assistant at Lomonosov MSU. There, I investigated the role of acid-sensitive two-pore potassium channels (TASK-1) in the vascular tone in rat arteries under varying pH conditions. This involved wire myography, RNA extraction, qPCR, and statistical analysis. Combined with my experience at the LABMGU Contract Research Organization, this experience trained me to work independently in the lab, collaborate across interdisciplinary teams, and communicate findings through both publication and presentation, while also strengthening my documentation and project management skills.

Motivated to gain international experience, I was awarded an Orange Tulip scholarship to study a Master's in Medical Biology at Radboud University. As a part of this program, I completed a 6-month research internship at the Donders Institute, where I quickly learned to operate a complex two-photon calcium imaging setup and successfully conducted *in vivo* experiments investigating social fear learning in mice. This experience enhanced my adaptability, precision, and capacity to handle technically demanding systems in a fast-paced environment.

While I have long been fascinated by the physiology of horses, I was unaware – having grown up in Russia – that equine sciences existed as a dedicated research field uniting molecular, cellular, and embryological approaches using equine models. Discovering this PhD opportunity was a revelation, as it integrates my passion for equine research with my ambition to build a career in translational science. I am especially enthusiastic about exploring post-zygotic aneuploidy using equine embryos due to the clear relevance to both veterinary and human reproductive medicine, as well as contributing to the development of non-invasive PGT systems for embryo screening. The integration of clinical activities and aneuploidy research, along with access to state-of-the-art imaging and sequencing technologies, makes this project both innovative and highly impactful.

Finally, I thrive when helping others and strongly value knowledge sharing. With experience mentoring first-year students and supporting them in understanding core biological concepts, I am very excited about contributing to teaching activities within this PhD. I also look forward to being part of the Utrecht Graduate School of Life Sciences, which I see as a valuable platform for both academic and professional development.

I am confident that my interdisciplinary training, practical lab experience, and dedication to research and education make me a strong candidate for this position. I would be honoured to join Utrecht University and contribute meaningfully to this project.

Thank you for considering my application. Should any questions arise, please do not hesitate to contact me.

Warm regards,
Varvara Lazarenko